



vulnerabilities consist primarily of flooding and, to a smaller extent, reduced water availability and erosion. The resulting water security⁹ challenges affect agricultural production and crop yields, thereby impacting livelihoods and food security.¹⁰

5. **Women are generally affected more adversely, both individually and as caretakers, by the impacts of climate change and natural disasters.** During droughts, for example, women and girls tend to eat less and spend more time collecting water.¹¹ It is estimated that women of childbearing age constitute 43.9 percent of the population potentially vulnerable to natural phenomena of very high recurrence.¹²

B. Sectoral and Institutional Context

Sectoral Context

6. **The agricultural sector represents approximately six percent of Peru's GDP and 16 percent of the country's total exports.**¹³ Almost a quarter of the total labor force, and more than half of the labor force in rural areas are employed in the agricultural sector (World Bank, 2022). The agriculture sector is also Peru's biggest water user, accounting for 89 percent of water withdrawals in the country.¹⁴

7. **Irrigation is a key driver of economic growth as it improves agricultural production, strengthens rural development, and increases resilience to climate change** by bridging the gap between crop water requirements and precipitation. Even though irrigated land represents only 36 percent (2.6 million hectares) of all agricultural land, approximately two-thirds of the country's agricultural output¹⁵ is produced on irrigated land. Moreover, irrigation could increase the country's GDP by 0.8 percent per year through agricultural productivity gains (World Bank, 2022).

8. **However, irrigation systems in Peru face a variety of challenges – low overall agricultural water use efficiency (WUE)¹⁶ – at only 30 percent,¹⁷ vulnerability of small-scale farmers, and low support capacity of local governments.**¹⁸ The low levels of WUE in irrigation systems are, in large part, due to the deteriorating condition of approximately 57 percent of Peru's existing irrigation infrastructure. Limited use of efficient on-farm irrigation technologies is also contributing to low water productivity, as they are not yet widely available or used throughout the country.

9. **More efficient on-farm irrigation systems¹⁹ enable farmers to use less water for their crops, thereby improving the overall irrigation performance, increasing agricultural water efficiency, and boosting agricultural productivity.** Only 13 percent of irrigated farmland employ modern on-farm water application methods (defined by the timely application of

⁹ Water security is defined as the availability of an acceptable quantity and quality of water for health, livelihoods, ecosystems, and production, coupled with an acceptable level of water-related risks to people, environments, and economies (Peru Water Security Diagnostic, 2022).

¹⁰ The FAO (2002) defines food security as 'a situation that exists when all people, at all times, have physical, social and economic access to sufficient, safe and nutritious food that meets their dietary needs and food preferences for an active and healthy life.' Peru has become the most food-insecure country in South America, with 51 per cent of the population is living in moderate food insecurity (FAO, 2021).

¹¹ The United Nations Framework Convention on Climate Change, 2022. Dimensions and examples of the gender-differentiated impacts of climate change, the role of women as agents of change, and opportunities for women.

¹² Ministerio de la Mujer y Poblaciones Vulnerables, 2016. "Convirtiendo los escenarios de desastre en oportunidades para la construcción de nuevos escenarios de igualdad. El MIMP ante el Fenómeno de El Niño". <https://www.mimp.gob.pe/files/direcciones/dcteg/Folleto-desastres-ctp.pdf>

¹³ World Bank Group. 2022. Peru Country Climate and Development Report. CCDR Series. World Bank, Washington, DC.

¹⁴ INEI, 2020, [Perú: Brechas de Género 2020 Avances hacia la igualdad de mujeres y hombres](#).

¹⁵ Agricultural output is defined as the total value of crops produced in economic terms, although it has also been used to denote (physical) crop yields.

¹⁶ Water use efficiency (WUE) is defined as the ratio between the volume of water used for irrigation and the volume extracted or derived from a water source for that purpose. It can be divided into storage, conveyance, distribution, and application efficiencies. It is measured by crop water consumption per water withdrawals. Water consumption refers to the amount of water that is depleted and thus unavailable for further use. Water withdrawal refers to the amount of water removed (or diverted) from a surface water or groundwater source.

¹⁷ The 30 percent overall irrigation water use efficiency in Peru is notably lower than those in Mexico (44 percent) and in Brazil (48 percent).

¹⁸ Source: Format No. 4-A, Gap Indicator "Percentage of agricultural land without technified irrigation", approved by Ministerial Resolution 0374-2021-MIDAGRI.

¹⁹ On-farm irrigation systems include drip or sprinkler systems. Drip systems lead water through a pipe system to the field, with drops of water delivered to plants at or near the root. In sprinkler systems, water is led to a field through a pipe system under pressure, and artificial rainfall is created to water the crops.